

Data Scientist and Machine Learning Engineer currently working in LLMs, NLP and some Computer Vision.

Employment History

Machine Learning Engineer Queensland University of Technology GenAI Lab (April 2024 - Present)	Machine Learning Engineer at Queensland University of Technology (QUT)'s newly established GenAI Lab. Technical responsibilities are varied, with deliverables ranging from simple learning materials in google colab notebooks, to scalable multi-container microservice architectures for LLM-powered API endpoints and multi-page websites with interactive educational resources. Additional responsibilities include workshop presentations, external stakeholder meetings on potential collaborations, learning material design, research consultation on the lower level mechanisms of generative AI, and contributions to research outputs. Work includes: ➤ GenAI Arcade - the flagship product from QUT's GenAI arcade – an educational resource intended for the general public, and used for educational outreach by the lab. I built and designed the website as well as most of the interactive “widgets” embedded into each page. Website built on Jekyll, hosted on GitHub Pages with custom domain, and most widgets built on Gradio and hosted on HuggingFace spaces, with plans to migrate to API calls for backend execution, and Svelte for UI. Viewable here. GitHub viewable here. ➤ BriX: Bridging via Exposure – an API built for the Center for Human-compatible AI pro-social ranking challenge. Targeting an AWS EC2 budget of \$5000 USD in hosting credits for 3 months and a p95 latency of 500ms, I built a dockerised multi-container micro-service architecture that used an off-the-shelf NLI model to estimate political polarity, and then reranked them using a diversification algorithm to reduce polarisation. FastAPI for API routing and load balancing, redis for database and task queue, celery for task workers. GitHub viewable here. ➤ Workshops – ran multiple workshops about Generative AI for the general public and media and communications researchers, from Retrieval-Augmented Generation and transformer architecture literacy to broader overviews of AI as a whole. Of note is RAGChat – a software built using streamlit, PyInstaller, ChromaDB and Ollama that allows users to create their own RAGs using their own documents. Code and executables viewable here. ➤ Tutorial series – wrote an interactive tutorial series on Stable Diffusion that covers each stage of the Stable Diffusion pipeline in great detail, including the history and/or theory of each stage. Focuses on Stable Diffusion 1.5, but the logic generalises to most modern image generation, as well as some video and audio generation. Written and available entirely in google colab.
Data Scientist Australian Taxation Office (Oct 2021 – March 2024)	Data scientist starting first in the UPEP program, before a year-long Graduate Program and ensuing full-time employment. I departed at the APS6 level after starting part-time at APS2. Work was predominantly focused on Natural Language Processing (NLP) and some Computer Vision in the Nexus and Document Understanding projects within Work Related Expenses substantiation. Work was done predominantly in Python, most used packages being numpy, pandas, PyTorch, Keras, scikit-learn, Dagster, spacy, and transformers. Some additional work also done in R and Teradata SQL. Work includes: ➤ Monitoring and debugging of complex, decoupled end-to-end Document Understanding pipeline, involving 11 PyTorch, Huggingface, Flair, and OpenCV2 models hosted on an AWS EC2 P2 instance. ➤ Maximizing efficiency of available hardware for model inference by identifying hardware bottlenecks, opportunities for low-precision compute and logical inefficiencies in the Python/SQL codebase. ➤ External consultation and business-wide insights on cloud infrastructure requirements for both large-scale deep-learning model pipelines and real-time inference for 1B+ parameter LLMs. ➤ Developed Proof-of-Concept multi-GPU asynchronous batch processing for inference. ➤ NLP data augmentation research and development using synonym swap and seq2seq models with Snorkel package.

Education

Master of Film and Television (Documentary) at Victorian College of Arts (2019 – 2021): GPA: 6.8/7.0. WAM: 80%. First Class Honours.	Studied documentary filmmaking, with an emphasis on the development of all technical and soft skills relevant to documentary filmmaking, alongside stylistic and artistic development. Developed transferrable skills in medium to large project and workflow management, internal and external stakeholder engagement, and novel and creative approaches to communication.
B.Arts/B.Science at Australian National University (2016 – 2018): GPA: 6.0/7.0. WAM: 75%	Studied a B.Arts/B.Sci double degree, majoring in Political Science, Economic Studies and Mathematical Economics, with a minor in Applied Statistics at the ANU. Primary focus and academic interest in Econometrics, Behavioural Economics and Game Theory, with some subjects in Computer Science, and advanced regression modelling and analysis.

Professional Skills

Online coursework:

- **Coursera - deeplearning.ai Deep Learning Specialisation:** A deep-learning course covering more advanced machine learning architectures such as deep neural networks, RNNs, CNNs, and Transformer based models.
- **Coursera – deeplearning.ai Natural Language Processing Specialisation (audited):** a course focused specifically on Natural Language Processing (NLP), covering topics like sentiment analysis, Word2Vec and GloVe embeddings, Named Entity Recognition, Siamese networks, and large language models like BERT and T5.
- **Udemy – Python for Data Science and Machine Learning Bootcamp:** a Python-specific machine learning bootcamp covering data exploration and visualization, supervised and unsupervised learning.
- **Teradata University:**
 - o **Introduction to Teradata Database**
 - o **Teradata SQL**
- **DataCamp:** Multiple short courses in Python and R at introductory and intermediate levels, covering tools and skills such as Tidyverse, Matplotlib, PyTorch, SQL, Shiny, Git, and Airflow.
- **Udemy – Motion Graphics & Data Visualization:** an entry-level introduction to data visualization and animation using vector graphics in Illustrator and After Effects.

- **Freecodecamp.org - Data Visualization with D3.js:** a free online course covering basic HTML and Javascript, focusing on responsive and animated svg creation for data visualization in D3.js
- **Freecodecamp.org – JavaScript Certification:** A free online course covering JavaScript concepts and syntax from basic to advanced.
- **Svelte.dev tutorial** – a short tutorial series on Svelte, for developing front-end/UI elements.

Other computer and software literacy:

- Proficient in SPSS, R, STATA and EViews for statistical and predictive modelling.
- Experience with Tableau and Power BI data visualization software
- Experience with Python for data science and deep learning: (Scikitlearn, Huggingface, PyTorch, TensorFlow, Keras, Seaborn, matplotlib, ggplot2)
- Experience with GitLab, GitHub and VSCode for git repository collaboration

Languages:

- English: first language
- Mandarin: basic
- French: basic